

Contract-grounded invoice auditing

Final Hospitals 2-5 submission | 17 September 2026 | Hospital 1 evaluation frozen

Scope, sequence and delivery

Started with Hospital 1's labelled development set, froze its engine/evaluation, then selected Hospital 4 for its simpler structured contract. At the user's final request, extended the same explicit calculation passes to Hospitals 2, 3 and 5. All 49,796 target line items and 3,968 header records were processed. Complete financial decisions remain bounded by evidence; unknown records are not accepted as correct. The six-to-eight-hour assessment cap is acknowledged, but prior human working time is unavailable, so no invented total-hours compliance claim is made.

Hospital	Submitted / unique IDs	Correct / erroneous	Coverage
2	492 / 1,125	481 / 11	43.73%
3	425 / 932	417 / 8	45.60%
4	374 / 835	364 / 10	44.79%
5	466 / 1,050	456 / 10	44.38%
Total	1,757 / 3,942	1,718 / 39	44.57%

The exact six-column submission contains both correct and erroneous opinions. The 2,185 omitted IDs retain overlapping reasons in outputs/final/omitted_invoices.csv. H4's previous 374 rows are unchanged; H2/3/5 add 1,383 rows. No target-hospital labels exist, so coverage is not accuracy. H1 is not submitted.

Grounding and calculation

Source-pinned references retain applicability, conditions, actions, scope, order, exact source text and uncertainty. H2 has 76 prose service clauses with embedded rules; reciprocal bundle rates are cross-checked. H3 has 118 original services, seven amended rates and two additions effective by service date from 2025-01-01. H5 has 84 services with separate facility and plan multiplier schedules. H4 retains its reviewed 98-service reference. Description matching uses inspected abbreviations and hospital-specific suffix removal; prices and billed units never choose a service. Acceptance thresholds are unchanged; no runtime model or experimental mappings are used.

Cross-invoice context precedes pricing. Original billed history, inferred delivered quantities and payable quantities remain separate. Rates follow bundle substitution, facility, plan, premiums/weekend uplifts, then deepest qualifying discount. Exact rational arithmetic rounds half away from zero to integer cents after each rate step, before quantity multiplication. H2/3/5 expressly define whole-term/all-patient volume scope; uncertain prior records widen bounds rather than disappear. H3 selects amended rates by service date, not invoice date. H5 applies and rounds facility and plan multipliers separately.

Assumptions and limits

The user approved H2's recorded service date as its Service Day because 07:00-boundary timestamps are absent, and H5's header facility code as line context because line-level codes are absent. These assumptions appear in evidence. H4 keeps approved inclusive same-patient bidirectional exclusions and earliest eligible lexical retention of exact copies; conflicting copies remain unresolved. H4 population/reset scope stays uncertain. Its three cap-corrected rows (000165/000540/000554) are explicitly assumption-dependent estimates, separated from confirmed cap violations.

H2/3/5 do not inherit H4's cap correction or duplicate allocation convention: affected amounts are withheld. H2 exclusions explicitly specify same patient and both directions; exact endpoints remain uncertain. H3/H5 positive exclusion cases are withheld because population/direction/endpoints are not explicit; same-patient candidate searching is a documented limitation. Wrong units and all dual-unit ambiguities remain unresolved. Operational compliance involving clinical records, actual submission timestamps, settlement or waivers cannot be established from invoice tables. H2's 60-day submission clause is screened without fabricating a submission date or automatic rejection.

Evaluation and reproducibility

Frozen Hospital 1 evidence

Original development: 717 IDs, 44/44 errors detected, zero invoice-level false positives, 398/717 decisions and 363/363 known expected totals exact. Historical held-aside: 196 IDs, 12/14 errors detected, zero false positives, 103/196 decisions and 93/94 known totals exact. Two errors remained under review. Labels are now exposed; no fresh-holdout claim or final-stage rule tuning is made. The grouped development category results below measure category identification; category false positives can occur on already-erroneous invoices.

Category	Development TP / FP / FN	Precision	Recall
Pricing	21 / 0 / 1	100%	95.45%
Arithmetic	9 / 5 / 0	64.29%	100%
Dates	10 / 5 / 0	66.67%	100%
Billing unit	8 / 1 / 0	88.89%	100%
Daily caps	3 / 0 / 0	100%	100%
Duplicate service	3 / 0 / 0	100%	100%
Duplicate invoice ID	5 / 0 / 0	100%	100%
Exclusions	2 / 0 / 1	100%	66.67%
Contract identity	3 / 0 / 0	100%	100%
Uncontracted service	0 / 0 / 10	Not measurable	0%

Four systematic failure types

Insufficient descriptions: missing qualifiers prevent defensible matching or unknown-service identification (H1-000657/000667). Shared uncertainty: unknown prior records affect later discounts and exclusions (000038/000045). Ambiguous records: repeated headers, malformed dates or wrong units block identity/quantity correction (000068/000148). Correction uncertainty: cap invoice 000015 was detected, but calculated 1,210,600 differs from labelled 1,195,825 by 14,775 cents. Exposed-label investigation of four cap cases suggests below-cap quantities; three residual reconstructions involve other unresolved lines, and the explanation is unproven. H4's estimates are approved, not validated; new hospitals' cap corrections remain withheld.

Numerical confidence

Keep the existing exposed-H1 class heuristic. Strict unique-ID rows with known amounts give 445/445 correct-class and 11/12 erroneous-class joint flag/exact-amount successes. Use 0.90 times the Wilson 95% lower bound, $z=1.96$: 0.892297 for correct and 0.581503 for erroneous rows. The 0.90 transfer factor is a declared heuristic. Correlated invoices, new contract interpretations and category errors limit interpretation. No target-hospital or cap-specific calibration, validated probability, or invented extra penalty is claimed. Submission evidence distinguishes context assumptions and H4 cap estimates from confirmed violations.

Verification and handover

103 tests cover extraction, original H1 behavior, amendment dates, new services, multipliers, per-step rounding, thresholds, cross-invoice bundles/exclusions, duplicate/cap safeguards, unit propagation and submission tampering. H2/3/5 manual checks replay 16 source-based line calculations, nine invoice deltas and nine uncertain cases; H4's existing 20 line calculations, ten erroneous deltas and boundary/cap checks are preserved. Example: H5-L00132-04 rounds 431,375 x1.05 to 452,944, then x.90 to 407,650 cents. Manual checks are not hidden-label validation.

Run `python3 -m assessment_audit`, then `python3 scripts/verify_assessment.py`. Python 3.9+ standard library suffices; PDF-only dependencies are pinned separately. Reproduction checks exact columns, identifiers, original billed cents, integer expected line sums, flags, confidence and output hashes. Source data and H1 remain hash-protected; previous H4 snapshots remain in Git history. Versioned prompts disclose AI-assisted development and approvals. README leads to the short evaluation report, one-page decision log, all evidence and omissions. Remaining work is evidence acquisition/adjudication and target confidence validation; no extra hospital coverage is claimed by guessing unresolved amounts.